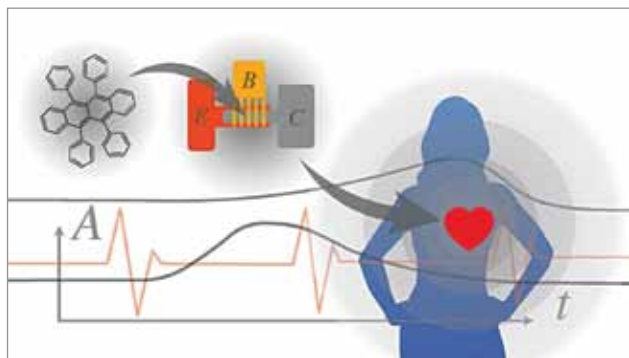


lithium-ion batteries, but their commercial use has been limited due to concerns over their potential for overheating and catching fire. This happens due to the production of dendrites—metallic structures that sprout as lithium accumulates on electrodes inside the battery. The mathematical model revealed that changing the electrolytes to ones with anisotropic qualities could reduce or even stop dendrite growth. Battery separators were another solution that was a huge aspect of the study.

Organic bipolar transistor much quicker than earlier organic transistors demonstrated

Transistors made of organic materials have gained attention recently for flexible applications. Now, the first extremely effective organic bipolar transistor has been demonstrated



Organic bipolar transistors can also handle demanding data processing and transmission tasks on flexible electronic elements, for example, here for electrocardiogram (ECG) data (Credit: TU Dresden)

by a research team at the Technical University of Dresden. The utilisation of highly organised thin organic layers was essential for this research. The working frequencies of the components have finally reached the gigahertz range, or more than a billion switching operations per second, making this new technology several times quicker than earlier organic transistors. Future applications such as intelligent patches with built-in sensors that process the data locally and wirelessly connect with the outside are envisaged.

This ink coating could replace batteries in wearable devices

Researchers at Sweden's KTH Royal Institute of Technology have developed a thermoelectric ink coating that can transform low-grade heat emitted by electronics into electrical power. When one end of a thermoelectric material is heated up, charge carriers (electrons and holes) travel away from the hot end and towards the cold end, resulting in an electric current. The team leveraged this phenomenon and

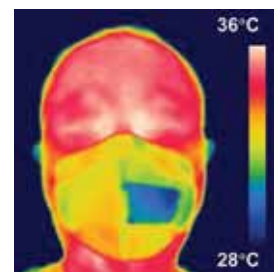


Hybrid thermoelectric material (Credit: KTH Royal Institute of Technology)

concentrated on the design and development of hybrid thermoelectric materials for room-temperature operations, which combine solid-state semiconductors with flexible materials like polymers to create inks. This technology could replace batteries by being integrated as a coating in wearable devices.

Bioelectronic face mask that could also monitor user's physiological state

The idea of putting bioelectronics into a breathable face mask has been put into practice by a research team at the University of Missouri College of Engineering. These masks can monitor a user's physiological state depending on the features of their cough. A technique called laser-assisted manufacturing that has been employed by researchers for about ten years has proved to be useful in the production of wearable bioelectronics. Laser-assisted fabrication is simple, scalable, cost-effective, and easily customisable. Also, the team looked at the viability of using the metallic conductor known as MoO₂, which exhibits properties that make it suitable for construction of various bioelectronic sensors.



An infrared image of a person wearing a smart mask concept (Credit: University of Missouri)

Micro-supercapacitors on marble flooring could provide excellent power supply

Eco-friendly construction materials, like marble and granite, are often used in construction and refurbishing. Now a group of scientists from the American Chemical Society (ACS), in a step toward integrating energy storage with these materials, have fabricated micro-supercapacitors onto the surface of stone tiles. However, these materials often have microscopic bumps and divots, making it difficult to adhere electrical components to them. Researchers have recently figured out how to place micro-supercapacitors, which have fast charging and discharging rates and excellent power supply storage, onto irregular surfaces with lasers. Scientists are adopting this approach to build micro-supercapacitors on marble.